

**Course: 24BSE2113-D**

Partial Differential Equations, Transforms & Optimization Techniques

**Group Assignment & Micro-Project-1**

**Phase 1: Theoretical Assignment**

*Topic 1: Quantum Confinement  
(The Schrödinger Wave Equation)*

**Group Number:** 13

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# 1 Introduction

Quantum confinement is one of the defining phenomena of modern nanoscale electronics. When the physical dimensions of a semiconductor structure shrink down to the scale of a few nanometers – comparable to the electron’s own de Broglie wavelength – the electron can no longer be treated as a free classical particle. Instead, it must be described as a matter wave whose behaviour is governed by quantum mechanics. This confinement forces the electron’s allowed energies to become discrete rather than continuous, which is the physical basis for technologies such as quantum dots, quantum well lasers, and modern transistor channels.

The simplest and most instructive model of this effect is the *particle in an infinite one-dimensional potential well*: an electron trapped between two infinitely high potential barriers a distance  $L$  apart. Despite its simplicity, this model captures the essential physics of confinement and provides a clean derivation of energy quantization using only the 1D Time-Dependent Schrödinger Equation, the method of separation of variables, and Fourier series representation.

This report develops that derivation step by step: starting from the governing partial differential equation, separating it into space and time parts, applying the physical boundary conditions of the well, extracting the quantized energy levels, normalizing the resulting eigenfunctions, and finally assembling the general time-dependent wave function as a Fourier series expansion.

## How Do We Solve a PDE Like This?

The Schrödinger equation is a *partial* differential equation (PDE) because  $\Psi$  changes with respect to two independent variables at once,  $x$  and  $t$ . Solving a PDE directly, with both variables tangled together, is difficult. The standard technique used throughout this project – *separation of variables* – gets around this by assuming the solution can be written as a product of two simpler, single-variable functions, one depending only on  $x$  and one depending only on  $t$ . Substituting this product back into the PDE splits it into two ordinary differential equations (ODEs), each of which is easy to solve on its own. The physical boundary conditions of the problem are then applied to the spatial ODE, and it is precisely this step that produces quantization. The rest of this report carries out that procedure in full detail.

### In Simple Terms

Think of it like solving one hard 2D puzzle by splitting it into two easy 1D puzzles: solve the "space" puzzle, solve the "time" puzzle separately, then multiply the two answers back together.

## 2 Objective

To analytically derive the quantized energy states and formulate the general wave function  $\Psi(x, t)$ , expressed as a Fourier series, for an electron trapped in a one-dimensional infinite potential well. The derivation is governed by the 1D Time-Dependent Schrödinger Equation and is obtained using the method of separation of variables.

## 3 Governing Equation and Setup

The state of a quantum particle is completely described by its wave function  $\Psi(x, t)$ , whose squared magnitude  $|\Psi(x, t)|^2$  gives the probability density of finding the particle at position  $x$  at time  $t$  [1]. The evolution of this wave function in time is governed by the 1D Time-Dependent Schrödinger Equation:

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} \quad (1)$$

Here  $\hbar$  is the reduced Planck constant,  $m$  is the mass of the particle (the electron, in this problem), and the right-hand side is the kinetic energy operator acting on  $\Psi$ .

For a particle confined to a 1D infinite potential well of length  $L$ , the potential energy function is

$$V(x) = \begin{cases} 0, & 0 \leq x \leq L \\ \infty, & \text{otherwise} \end{cases} \quad (2)$$

Inside the well the particle is completely free (no force acts on it), while outside the well the potential is infinite, meaning it would take infinite energy for the particle to be found there. Physically, this means the probability of finding the particle outside the well is exactly zero, and by continuity of the wave function this forces  $\Psi$  itself to vanish at the walls. This gives the Dirichlet boundary conditions:

$$\Psi(0, t) = 0 \quad \text{and} \quad \Psi(L, t) = 0 \quad \forall t \geq 0 \quad (3)$$

These two conditions are what will later force the electron's energy to take only discrete, quantized values.

### In Simple Terms

The electron is trapped in a box and can never be found outside it. Since the walls are "infinitely hard," the wave function has to fade to exactly zero right at the edges of the box – there is simply no way for the particle to leak through.

## 4 Method of Separation of Variables

Equation (1) is a linear partial differential equation in two independent variables,  $x$  and  $t$ . A standard and powerful technique for solving such linear PDEs – and the one required by this assignment – is separation of variables [2], where we look for solutions that factor into a purely spatial part and a purely temporal part:

$$\Psi(x, t) = \psi(x) \phi(t) \quad (4)$$

This assumption is justified because the boundary conditions in Equation (3) apply only at fixed positions ( $x = 0, L$ ) for all time, which is exactly the situation in which a separable, or “stationary state,” solution exists.

Substituting Equation (4) into Equation (1) and using the product rule (noting that  $\partial\Psi/\partial t = \psi(x)\phi'(t)$  and  $\partial^2\Psi/\partial x^2 = \psi''(x)\phi(t)$ ):

$$i\hbar\psi(x)\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\phi(t)\frac{d^2\psi(x)}{dx^2} \quad (5)$$

Dividing both sides of Equation (5) by the product  $\psi(x)\phi(t)$  separates every term so that all  $t$ -dependence sits on the left and all  $x$ -dependence sits on the right:

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} \quad (6)$$

The left-hand side of Equation (6) is a function of  $t$  only, while the right-hand side is a function of  $x$  only. Since  $x$  and  $t$  are independent variables, the only way an expression in  $t$  alone can equal an expression in  $x$  alone, for all  $x$  and  $t$  simultaneously, is if both sides are equal to one and the same constant. We call this constant  $E$ , which will turn out to be the particle’s total energy (its eigenvalue):

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} = E \quad (7)$$

This is the elimination-of-variables step: the single PDE in Equation (1) has now been converted into two much simpler ordinary differential equations (ODEs) – one purely in  $t$  and one purely in  $x$  – linked together only by the shared constant  $E$ .

### In Simple Terms

After dividing through, one side of the equation only knows about time and the other side only knows about position. Two completely unrelated quantities can only stay equal to each other, no matter what values  $x$  and  $t$  take, if they are both secretly the same fixed number. That number is  $E$  – the particle’s energy.

## 5 Temporal Solution

Isolating the time-dependent equality from Equation (7):

$$i\hbar\frac{d\phi(t)}{dt} = E\phi(t) \implies \frac{d\phi(t)}{dt} = -\frac{iE}{\hbar}\phi(t) \quad (8)$$

Equation (8) is a first-order linear ODE of the standard form  $\phi'(t) = \lambda\phi(t)$ , whose solution is a simple exponential. Solving it gives

$$\phi(t) = C \exp\left(-\frac{iEt}{\hbar}\right) \quad (9)$$

where  $C$  is an integration constant that will later be absorbed into the normalization of the spatial part. Because the exponent is purely imaginary,  $\phi(t)$  has constant magnitude  $|C|$  and only rotates in phase as time progresses – this is why solutions of this form are

called *stationary states*: the probability density  $|\Psi(x, t)|^2 = |\psi(x)|^2 |\phi(t)|^2 = |\psi(x)|^2 |C|^2$  does not change with time, even though the wave function's phase does.

### In Simple Terms

Picture a clock hand spinning at a steady rate. Its length never changes – only the angle it points at changes. In the same way,  $\phi(t)$  never grows or shrinks, it just "spins" in phase, which is why the shape of where the electron is likely to be found stays fixed over time even though the wave function itself is still evolving.

## 6 Spatial Solution and Boundary Conditions

Isolating the space-dependent equality from Equation (7):

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E \psi(x) \implies \frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \quad k = \frac{\sqrt{2mE}}{\hbar} \quad (10)$$

Equation (10) has exactly the same form as the classical simple-harmonic-oscillator equation, and so its general solution is a combination of sine and cosine:

$$\psi(x) = A \sin(kx) + B \cos(kx) \quad (11)$$

The constant  $k$  here plays the role of a spatial wavenumber, related to the particle's momentum by  $p = \hbar k$ .

### Applying the Boundary Conditions

We now use the physical boundary conditions from Equation (3) to pin down the constants  $A$ ,  $B$ , and the allowed values of  $k$ .

**At  $x = 0$ :**

$$\psi(0) = A \sin(0) + B \cos(0) = A(0) + B(1) = B = 0 \quad (12)$$

Since  $\sin(0) = 0$  and  $\cos(0) = 1$ , Equation (12) forces  $B = 0$  directly, leaving the simplified spatial solution  $\psi(x) = A \sin(kx)$ .

**At  $x = L$ :**

$$\psi(L) = A \sin(kL) = 0 \quad (13)$$

There are two ways to satisfy Equation (13): either  $A = 0$ , which is the trivial (physically meaningless, "no particle at all") solution, or  $\sin(kL) = 0$  with  $A \neq 0$ . We keep the non-trivial option, which requires

$$\sin(kL) = 0 \implies k_n L = n\pi, \quad n = 1, 2, 3, \dots \quad (14)$$

(the case  $n = 0$  is excluded because it forces  $\psi(x) \equiv 0$  everywhere, again describing no particle). Solving Equation (14) for  $k$  gives the allowed, discrete wavenumbers:

$$\boxed{k_n = \frac{n\pi}{L}} \quad (15)$$

This is the heart of quantum confinement: the boundary conditions permit only standing waves for which a whole number of half-wavelengths fit exactly inside the well, exactly analogous to the allowed standing wave modes on a guitar string fixed at both ends.

**In Simple Terms**

Clamp both ends of a guitar string and pluck it – it can only vibrate at certain fixed frequencies (and their overtones), never at an arbitrary frequency, because only whole numbers of half-waves can fit between the two fixed ends. The electron in a box obeys exactly the same geometric rule, just applied to a probability wave instead of a physical string.

## 7 Derivation of Quantized Energy States

Having restricted  $k$  to the discrete values  $k_n$ , we now translate this restriction back into a restriction on energy. Substituting  $k_n = \frac{n\pi}{L}$  into the definition  $k = \frac{\sqrt{2mE}}{\hbar}$  from Equation (10):

$$\frac{n\pi}{L} = \frac{\sqrt{2mE_n}}{\hbar} \quad (16)$$

Squaring both sides of Equation (16) removes the square root:

$$\frac{n^2\pi^2}{L^2} = \frac{2mE_n}{\hbar^2} \quad (17)$$

and solving for  $E_n$  gives the quantized energy levels of the particle in a box:

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots \quad (18)$$

This is a striking departure from classical mechanics: a classical particle bouncing between two walls could have any energy whatsoever, but a quantum particle confined the same way can only occupy this discrete “ladder” of energies [3], spaced further and further apart as  $n$  increases (since  $E_n \propto n^2$ ). Note also that  $E_n$  grows as  $1/L^2$  – the more tightly the electron is confined, the larger the spacing between its allowed energy levels becomes. This inverse-square dependence on confinement length is precisely why quantum confinement effects become significant only at the nanoscale.

**In Simple Terms**

A classical ball bouncing between two walls can roll at any speed you like, so it can have any energy. An electron trapped the same way cannot – it is only allowed to sit on specific “rungs” of an energy ladder, never in between, purely because of the wave-fitting requirement derived above.

## 8 Normalization of the Spatial Eigenmodes

The wave function must satisfy one more physical requirement: since  $|\psi_n(x)|^2$  represents a probability density, the total probability of finding the particle *somewhere* inside the well must equal exactly 1. This normalization condition fixes the remaining constant  $A$ :

$$\int_0^L |\psi_n(x)|^2 dx = 1 \quad \implies \quad A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1 \quad (19)$$

To evaluate the integral in Equation (19), we use the half-angle identity  $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ :

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \int_0^L \frac{1}{2} \left[ 1 - \cos\left(\frac{2n\pi x}{L}\right) \right] dx = \frac{L}{2} - \underbrace{\frac{L}{4n\pi} \sin\left(\frac{2n\pi x}{L}\right) \Big|_0^L}_{=0} = \frac{L}{2} \quad (20)$$

The cosine term integrates to a sine that vanishes at both  $x = 0$  and  $x = L$  for every integer  $n$ , leaving simply  $L/2$ . Substituting this back into Equation (19):

$$A^2 \cdot \frac{L}{2} = 1 \quad \implies \quad A = \sqrt{\frac{2}{L}} \quad (21)$$

Hence the normalized spatial eigenfunctions are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (22)$$

These functions form an orthonormal set on  $[0, L]$ , meaning  $\int_0^L \psi_n(x) \psi_m(x) dx = 0$  for  $n \neq m$  and  $= 1$  for  $n = m$  – a property that will be used directly in the next section to extract the Fourier coefficients.

### In Simple Terms

This step has nothing to do with quantization itself – it is just bookkeeping. We are simply scaling the wave so that "total probability of finding the electron somewhere in the box" comes out to exactly 100%, the way any probability should.

## 9 Final Fourier Series Solution

Combining the normalized spatial eigenfunction, Equation (22), with the temporal solution, Equation (9) (absorbing the constant  $C$  into  $A$ ), gives the complete stationary-state (standing-wave) solution for a single mode  $n$ :

$$\Psi_n(x, t) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \exp\left(-i \frac{E_n t}{\hbar}\right) \quad (23)$$

Each  $\Psi_n(x, t)$  individually satisfies the Schrödinger equation and both boundary conditions. However, the electron is not restricted to occupying only one of these modes; because Equation (1) is linear, *any* linear combination of solutions is itself a solution. This is the superposition principle, and it lets us build the most general possible wave function as a sum over all the allowed modes – precisely a Fourier series in the spatial sine-basis  $\{\psi_n(x)\}$  [2]:

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \exp\left(-i \frac{n^2 \pi^2 \hbar}{2mL^2} t\right) \quad (24)$$

### In Simple Terms

This is exactly the same idea as breaking a complicated sound wave into a mix of pure musical tones (a standard Fourier series). Here, the "pure tones" are the box's own allowed standing-wave modes  $\psi_n(x)$ , and  $c_n$  tells you how much of each pure mode is present in the electron's actual state.

The Fourier coefficients  $c_n$  are not free parameters; they are completely determined by the particle's initial wave function  $\Psi(x, 0)$  at  $t = 0$ . Setting  $t = 0$  in Equation (24) gives  $\Psi(x, 0) = \sum_n c_n \psi_n(x)$ , and because the  $\psi_n(x)$  are orthonormal, we can project out any single coefficient by multiplying both sides by  $\psi_n(x)$  and integrating over  $[0, L]$ :

$$c_n = \int_0^L \Psi(x, 0) \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) dx \quad (25)$$

### Physical Interpretation

Equation (24) states that *any* allowed wave function inside the box, however complicated, can be written as a weighted sum of the discrete standing-wave modes  $\psi_n(x)$ , each one oscillating in time at its own characteristic frequency  $E_n/\hbar$ . The weights  $c_n$  are set entirely by the initial condition through Equation (25). This is exactly the same eigenfunction-expansion idea used to build an ordinary Fourier series on an interval – only here the "basis" functions are not a generic sine/cosine set but the specific quantized eigenmodes forced upon the system by the physical boundaries of the potential well.

## 10 Conclusion

Starting from the 1D Time-Dependent Schrödinger Equation, this report has shown, step by step, how confining an electron to a region of length  $L$  with infinitely high walls leads directly to quantization. Separation of variables split the governing PDE into an exponential temporal equation and a harmonic-oscillator-type spatial equation. Applying the physical boundary conditions  $\Psi(0, t) = \Psi(L, t) = 0$  restricted the spatial solutions to a discrete family of sine modes,  $\psi_n(x) = \sqrt{2/L} \sin(n\pi x/L)$ , and correspondingly restricted the particle's energy to the discrete ladder  $E_n = n^2\pi^2\hbar^2/(2mL^2)$ .

Combining the normalized spatial modes with their time evolution and invoking the superposition principle produced the general solution as a Fourier series, Equation (24), with coefficients  $c_n$  fixed by the initial state via Equation (25). Two engineering conclusions follow directly from this derivation. First, confinement is what *creates* discreteness – a free electron ( $L \rightarrow \infty$ ) has a continuous energy spectrum, but as  $L$  shrinks the energy levels spread apart as  $1/L^2$ , which is precisely the mechanism exploited in quantum dots, quantum wells, and nanoscale transistor channels to engineer specific, tunable electronic and optical properties. Second, the Fourier series structure of the general solution shows that the time evolution of *any* initial electron state in such a structure can be predicted exactly, simply by decomposing it into these quantized eigenmodes – a technique that extends naturally to more realistic 2D and 3D confinement geometries used in real semiconductor device design.

## References

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